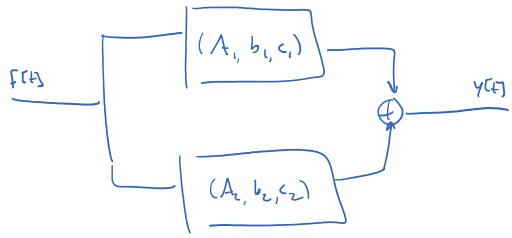
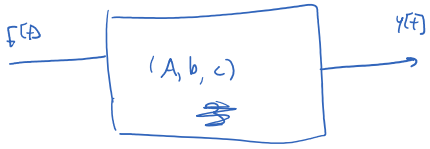


$f_1 = f_2$

$$x_1(t+1) = A_1 x_1 + b_1 f_1$$
$$y_1(t) = c_1^T x_1$$

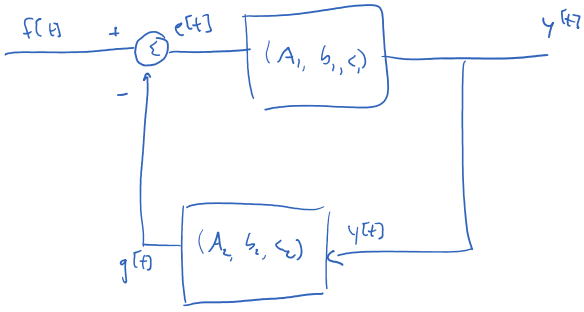


$$x_2(t+1) = A_2 x_2 + b_2 f_2$$
$$y_2(t) = c_2^T x_2$$

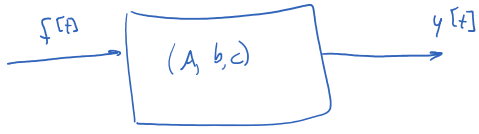
$$y[t] = y_1 + y_2$$
$$= c_1^T x_1 + c_2^T x_2$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

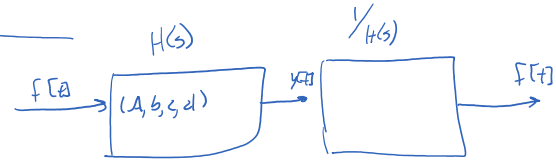
$$\boxed{\begin{aligned} y[t] &= \begin{bmatrix} c_1^T & c_2^T \end{bmatrix} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x} \\ X(t+1) &= \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} f \end{aligned}}$$



$e[t] = f[t] - g[t]$



(A, b, c, d)



$x[t+1] = A x[t] + b f[t]$

$y[t] = c^T x[t] + d f[t]$

$$x[k+1] = A x[k] + b f[k]$$

$$y[k] = c^T x[k] + d f[k]$$

$$f[k] = \frac{1}{d}\left(y[k] - c^T x[k] \right)$$

$$x[k+1] = A x[k] + \frac{1}{d} b \left(y[k] - c^T x[k] \right)$$

orig: $H(z) = c^T (zI - A)^{-1} b + d$

$$A \longrightarrow T^{-1} A T$$

$$(zI - A)^{-1} \longrightarrow (zI - T^{-1} A T)^{-1}$$

$$= (z T^{-1} T - T^{-1} A T)^{-1}$$

$$\begin{bmatrix} A & A_1 \\ 0 & A_2 \end{bmatrix}^{-1}$$

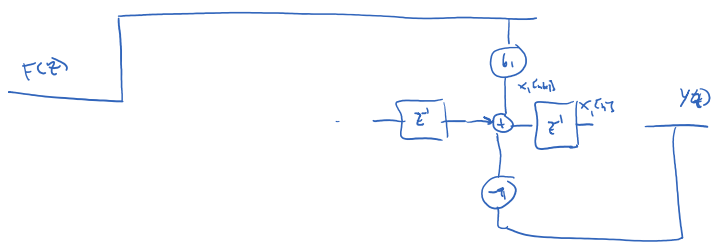
$$\left(zI - \begin{bmatrix} A & A_1 \\ 0 & A_2 \end{bmatrix} \right)^{-1}$$

$$y = x + w$$

Gaussian Gaussian

$$f(x|y) \sim N(\mu_{x|y}, R_{x|y})$$

$$= \frac{1}{(2\pi)^{\frac{d}{2}} |\det(R_{x|y})|^{\frac{1}{2}}} \exp\left[-\frac{1}{2} (x - \mu_{x|y})^T R_{x|y}^{-1} (x - \mu_{x|y}) \right]$$



$$x_1[k+1] = b_1 f[k] - a_1 y[k] + x_2[k]$$

$$c = [1 \quad 5 \quad 6] \qquad s^2 + 5s + 6$$

roots(c)

show: $A(j\omega) = \frac{1}{b}\sqrt{(b-\omega^2)^2 + (a\omega)^2}$ $\tan\phi(j\omega) = \frac{a\omega}{b-\omega^2}$

$A(j\omega)\, b = \sqrt{(b-\omega^2)^2 + (a\omega)^2}$

$$\begin{bmatrix} A(j\omega_1) & -\omega_1\sqrt{1+\frac{1}{\tan^2\phi(j\omega_1)}} \\ & \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix}$$